

Scaling for Success?

Coverage, Timing, and the Reach of Canadian Federal Innovation Support

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Abstract

This paper provides a descriptive account of how scale-up firms interact with Canada’s federal innovation policy mix. Linking Statistics Canada firm records with the Business Innovation and Growth Support (BIGS) database and Scientific Research and Experimental Development (SR&ED) tax credit records for 2007–2019, we show which firms scale and when they receive federal support. Few in number, scale-ups account for a disproportionate share of employment and revenue, yet most never receive federal support. SR&ED typically reaches firms before or during scaling, whereas BIGS funding is taken, on average, afterward. Among matched scale-ups, recipients show stronger subsequent employment, revenue, R&D, wage, and export trajectories than comparable non-recipients, although productivity differences are insignificant. These patterns are suggestive of a causal contribution, although the design cannot rule out selection on unobserved firm characteristics. Overall, federal innovation policy appears to reinforce firms already on a growth trajectory more often than it catalyzes early-stage scaling.

Keywords: scale-ups, high-growth firms, innovation policy, administrative data, Canada.

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1 Introduction

It is well established that a relatively small share of firms accounts for a disproportionate share of net job creation, innovation, and aggregate output growth in advanced industrial economies (Birch, 1979; Birch et al., 1995; Henrekson and Johansson, 2010; Nesta, 2009; Coad et al., 2014). These high-growth firms, commonly referred to as scale-ups, are as important in Canada as they are elsewhere. While most firms grow slowly, if at all, a small cohort of firms are drivers of economic dynamism (Dixon and Rollin, 2014). Accordingly, Canadian policymakers have sought to reorient federal innovation and growth policies to support firms with such potential (Innovation, Science and Economic Development Canada, 2019a,b).

The Canadian innovation policy mix combines an indirect instrument (the Scientific Research and Experimental Development, or SR&ED, tax credit) with a wide array of direct-spending programs delivered through the Industrial Research Assistance Program (IRAP), the Regional Development Agencies (RDAs), the Strategic Innovation Fund (SIF), the Trade Commissioner Service (TCS), and others. These programs differ in eligibility, in the firm populations they reach, and in the stage at which firms typically engage with them.

Until recently, it was hard to assess whether and how that policy support reaches scale-ups. The administrative data records that we need to observe firm growth and participation across these support instruments were not collectively available. Doing so required linking firm-level administrative records on growth and performance to direct and indirect support program receipt records. We can now do this. Using Statistics Canada administrative firm records linked with the new Business Innovation and Growth Support (BIGS) database and existing SR&ED program records, we examine federal innovation support at the firm level with a focus on scale-ups. To our knowledge, this is the first study to use these sources jointly for the scale-up firm population.

A recent Statistics Canada report documents patterns of overlap between BIGS and

SR&ED participation among Canadian firms ([Statistics Canada, 2024](#)). We extend this evidence by focusing specifically on scale-ups, examining the timing of federal support relative to growth trajectories, and then linking support use to subsequent firm-level performance. Similarly, while [Dowsett and Kelly \(2025\)](#) analyze firm participation in federal innovation programs, they do not focus on the scale-up population or examine the sequencing of BIGS and SR&ED support relative to firms' growth trajectories. Our contribution brings all of these elements together for a firm-level analysis of how high-growth firms engage with federal innovation supports before, during, and after scaling.

We examine Canadian firms over the 2007–2019 period, classifying them as scale-ups under two definitions: the OECD employment definition, defined as firms with 10 or more employees achieving annualized average employment growth above 20 percent over three years, and a revenue-based definition adapted from the Kauffman Foundation, which identifies firms with at least \$2 million in annual revenue by the end of a three-year period and annualized revenue growth of at least 20 percent over that period. The scale-up population identified for analysis consists of firms that meet at least one of these definitions. We do not consider the full population of Canadian firms.

We ask four questions:

1. Among Canadian firms, who scales up, and what is the industrial and geographical distribution?
2. Among scale-ups, what share receives federal innovation policy support in the form of SR&ED and BIGS?
3. When, during a firm's growth trajectory, are federal program supports used, and how do indirect (SR&ED) and direct (BIGS) instruments differ in their timing?
4. Among scale-ups, do program recipients show different subsequent employment, revenue, R&D, wage, and export trajectories than similar non-recipients?

The fourth question relates to a long-running policy debate about whether Canadian

innovation programs reach the firms they intend to reach and whether support is associated with subsequent growth and innovation (Nicholson, 2018; Stewart, 2025). Using a propensity-score-matched comparison and firm-level fixed-effects regression on the matched sample, we estimate conditional associations, acknowledging that selection on unobserved characteristics that vary across firms (e.g., human capital or management structure) cannot be addressed given our design. We therefore report conditional descriptive associations and interpret them as such.

The findings, then, are as follows. Consistent with expectations based on prior research, scale-ups represent a small share of Canadian firms (6–7 percent under the OECD employment definition; 1.5–2 percent under the Kauffman revenue definition) but account for a disproportionate share of employment and revenue. Geographically, they concentrate in Canada’s largest urban Economic Regions, including Toronto, Vancouver (Lower Mainland–Southwest), Montréal, Calgary, Edmonton, Ottawa, and Kitchener–Waterloo–Barrie. Within-region scale-up shares are broadly comparable across the country under the OECD employment definition.

We find that most scale-ups never receive federal innovation support. Among those who do, SR&ED tends to reach firms before or during scaling and BIGS support, on average, is provided to firms after they have scaled. Scale-ups that receive support subsequently exhibit stronger performance across standard outcome measures than comparable non-recipients, a finding subject to the limitations inherent to a design that selects on observables.

The remainder of the paper proceeds as follows. Section 2 reviews the relevant scale-up literature and Canadian policy context. Section 3 describes the data linkage and the empirical design, including the limitations of the matching approach. Section 4 presents the descriptive findings. Section 5 discusses the implications for Canadian innovation policy. Section 6 concludes.

2 Scale-Ups and Canadian Innovation Policy: Brief Context

What is a scale-up? Definitions vary and the choice of definition will determine which firms are identified (Petersen and Ahmad, 2007; Daunfeldt et al., 2014; Coad et al., 2014). The OECD’s widely used definition classifies a high-growth enterprise as a firm with at least 10 employees at the start of a three-year window achieving average annualized growth above 20 percent in employees or revenue (Ahmad, 2006; Eurostat and OECD, 2008). The Kauffman Foundation defines growth entrepreneurship partly by firms reaching specified revenue or employment thresholds within a fixed time after founding (Morelix et al., 2016). Other work defines high-growth firms as the top decile by growth or by absolute employment increase (Clayton et al., 2013; Birch, 1979; Schreyer, 2000). Employment-based and revenue-based definitions often identify different dimensions of growth and thus different kinds of firms (Daunfeldt et al., 2014), which is why we find it important to use definitions that capture employment and revenue growth.

Why do scale-ups matter? Firm growth in advanced economies is highly skewed. Most firms grow little, while a small minority grows very fast (Birch, 1979; Henrekson and Johansson, 2010; Nesta, 2013; Anyadike-Danes et al., 2015). The small share of firms that grow rapidly accounts for a large share of net job creation (Picot and Dupuy, 1998; Haltiwanger et al., 2013; Coad et al., 2014) and contribute to aggregate productivity and output growth (Haltiwanger et al., 2017). Scale-ups disproportionately innovate (Coad and Rao, 2008), export (Love and Roper, 2015), and operate in dynamic regional clusters (Bramwell and Wolfe, 2016; Spencer et al., 2010; Denney et al., 2021a; Creutzberg et al., 2024).

Despite their importance, Canadian scale-ups have traditionally received less policy attention than other drivers of economic growth. Historically, Canada’s growth strategy focused on large, often foreign-owned, resource-based, or manufacturing firms rather than fast-growing smaller firms in, for example, the technology sector. Innovation policy has typically emphasized multinational firms and relied on technology imported from the US through the

presence of MNEs (Britton, 2003; Nicholson, 2016; Wolfe et al., 2022), overlooking the potential contribution of homegrown firms (Wolfe, 2015; Denney et al., 2023). However, scale-up and entrepreneurial firms have recently gained prominence in policy discourse, highlighted by federal investments in 2019 explicitly aimed at scaling up Canadian companies as part of a broader innovation agenda (Innovation, Science and Economic Development Canada, 2019a).

Canada’s innovation policy support combines indirect support for business R&D, principally through the Scientific Research and Experimental Development (SR&ED) tax credit, with direct support delivered through a wide range of federal programs (Denney et al., 2023; Stewart, 2025). In our data, these direct-support programs are captured in Statistics Canada’s Business Innovation and Growth Support (BIGS) database, which brings together firm-level information on participation in and funding from federal business innovation and growth programs. These include the Industrial Research Assistance Program (IRAP), innovation programming delivered through the Regional Development Agencies (RDAs), the Strategic Innovation Fund (SIF), the Trade Commissioner Service (TCS), and others. SR&ED, by contrast, provides tax credits for qualifying R&D expenditures and is refundable for Canadian-controlled private corporations under specified thresholds.

By aggregate dollar volume, SR&ED is Canada’s largest single innovation policy instrument. Public support for business R&D in Canada, much of it delivered indirectly, has been high by OECD standards; private R&D spending and the entrepreneurship rate, by contrast, have lagged peers, while policy attention to homegrown scale-ups is comparatively recent (Council of Canadian Academies, 2018, 2025; Stewart, 2025).

This mix of indirect and direct support also distinguishes Canada from peer economies. By comparison, the U.S. system places greater emphasis on direct support through the Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs. These initiatives provide competitive R&D funding through federal agencies, linking support for small firms to agency-specific research priorities and commercialization

objectives.

The instruments also differ regarding the firms they are designed to reach and the way firms can use them. SR&ED is available to firms that incur qualifying R&D expenditures, whereas many of the direct-support programs require firms to apply and undergo some form of assessment. These differences may affect both which scale-ups receive support and when in their growth trajectories they receive it. We return to this distinction, which we consider important, in Section 4.

Despite the relatively high level of public support provided, the failure of the existing policy mix to arrest Canada’s downward innovation trajectory represents a continuing policy paradox (Nicholson, 2018; Council of Canadian Academies, 2018, 2025; Stewart, 2025). For the present study, a more specific question is how that policy mix intersects with the growth trajectories of Canadian scale-up firms. Research by Denney et al. (2023) suggests that Canadian scale-up entrepreneurs perceive gaps in the policy mix, including a need for more targeted support and growth-stage financing rather than generic SME measures. They emphasize constraints on capital, talent, and market access as barriers to growth.

How, then, does the federal innovation support system engage firms that achieve high-growth status? This remains an important but largely unanswered question. Answering it requires knowing not only whether scale-ups receive federal support, but which forms of support they access and when that support occurs relative to scaling. Existing firm-level evidence provides important baseline information on participation in federal innovation programs across the broader Canadian firm population, but we know much less about scale-ups specifically. By examining SR&ED and BIGS jointly for this population, we provide a novel descriptive account of who receives federal innovation support and when in the scaling process they receive it.

3 Data and Methods

We make use of several Statistics Canada administrative firm-level datasets, including the Business Register, Payroll Deduction records, Exporter Register, and Corporate Income Tax files. These data cover the universe of businesses operating in Canada and provide detailed information on firm characteristics and performance, including revenue, employment, exports, R&D activity, profits, age, and industry. We also link the administrative data to PATSTAT patent records, which provide information on patent applications, and to firm-level records on federal innovation support. Federal supports include (1) the BIGS database, which captures firm-year participation and funding amounts across major federal direct-support innovation programs; and (2) SR&ED program records, which provide information on qualifying R&D expenditures and tax credit claims. The linked dataset spans the period 2007–2019.

Moreover, we draw on two versions of the linked data with different coverage periods. The benchmarking results for scale-ups use an earlier version covering 2007–2016 ([Denney et al., 2021b](#)), while the examination of innovation support among scale-ups uses a later version extending to 2019. Both are based on the same underlying data sources and linkage framework.

We identify scale-ups using two definitions. One is the OECD employment measure, which requires firms to have at least 10 employees at the beginning of a three-year period and annualized employment growth above 20 percent over that period. The second is a Kauffman Foundation measurement that uses revenue. To meet this definition, firms need to reach at least \$2 million in annual revenue by the end of a three-year period while achieving annualized revenue growth of at least 20 percent. The two measures identify partially overlapping but distinct scale-up firm populations defined by different dimensions of growth.

For each type of scale-up, we report unweighted shares, counts, and means for the scale-up population and for relevant subsets, including industry, Economic Region, size class,

technology intensity, exporter status, R&D performer status, and patent-holder status.

To examine the relationship between federal support and subsequent firm performance, we estimate firm-level fixed-effects regressions on a propensity-score-matched sample. For each scale-up firm that first receives BIGS support in a given calendar year (i.e., the treated firms), we identify the set of scale-ups that do not receive BIGS support in that year as the candidate control pool. We estimate the probability of treatment as a function of pre-treatment covariates based on industry (two-digit NAICS), size class, firm age, R&D performer status, exporter status, patent-holder status, and lagged values of the outcome variables. We match each treated firm to its nearest neighbors on the estimated propensity score and retain the matched sample for the second-stage regression:

$$Y_{it} = \alpha + \beta \text{BIGS}_{it} + X'_{it}\gamma + \mu_i + \theta_{j(i),t} + \varepsilon_{it}, \quad (1)$$

where Y_{it} is the outcome (asinh-transformed employment, revenue, average wages, R&D expenditure, or exports; or the linear-probability dependent variable for scaling incidence, R&D incidence, and export incidence) for firm i in year t ; BIGS_{it} is an indicator equal to one from the year of first BIGS receipt and one thereafter; X_{it} contains controls including SR&ED support; μ_i is a firm fixed effect; and $\theta_{j(i),t}$ is an industry-by-year interaction. The sample spans $t - 5$ to $t + 5$ around treatment. Standard errors are clustered at the firm level.

For SR&ED-as-treatment estimates we report pooled-OLS estimates with lagged outcome, firm age, industry, province, and year fixed effects, using the full firm-year panel rather than a matched cohort. Notably, the matched-cohort design is a propensity-score-matching procedure, not an identification strategy. We discuss the specification difference and what it means for interpretation in more detail in Section 4, and address some design limitations here.

The first limitation concerns the adequacy of the covariate set. The propensity-score model uses variables available in the linked administrative data. Our chosen set of variables is comparable to those used in published evaluations of similar programs in other juris-

dictions (Howell, 2017; Santoleri et al., 2024). Including lagged outcomes helps account for persistence in firm performance by balancing treated and control firms on their pre-treatment characteristics and trajectories. Because those lags enter the matching model, part of any estimated post-treatment difference may reflect persistence in the outcomes themselves rather than program participation. However, the administrative data, while highly detailed, are still observational. They do not capture several factors that may affect both firm growth and program participation, including but not limited to the human capital of founders and managers, the quality and commercial potential of intellectual property, and other organizational capabilities. Nor do we have a research design (e.g., a discontinuity in program eligibility or another source of exogenous variation) that would let us recover a causal effect from observational data of this kind.

The data and design limits are real, but we also think our particular sample mitigates some concerns by restricting both the treatment and control groups to firms that have already achieved scale-up status. Treatment is defined as the first receipt of innovation support after scaling up, while the control group sample is drawn from peer scale-ups that did not receive support. Using this particular cohort of firms (i.e., those having already experienced scale-up growth) in theory reduces much of the heterogeneity in firm and human capital quality that arises when supported firms are compared with the full population. The estimates should therefore be interpreted as differences among high-performing firms only.

Even within this more comparable group, unobserved selection remains possible. Firms with stronger management teams, better and more developed growth strategies, higher-quality technologies, and other unobservable or intangible qualities may be more likely both to obtain federal support and to sustain strong performance without it. Furthermore, many BIGS-supported programs require firms to apply and then pass an adjudication process that rewards commercial readiness and growth potential. Because these characteristics are difficult to measure in administrative data, the matching specification cannot fully account for them.

The bias “left over” even after limiting the firm selection pool to scale-ups is therefore likely upward. If firms receiving support are structurally and systematically stronger than otherwise similar scale-ups along dimensions that are unobserved to the researcher, part of the estimated post-treatment performance differential may reflect those underlying advantages rather than the causal effect of support itself. Consequently, the results should be interpreted as evidence of an association between support receipt and subsequent performance among scale-ups, suggestive of causal effects (and thus potential areas for future research) but not definitive causal estimates.

In light of these limitations, the matched-firm estimates should be read as conditional descriptive associations under stated assumptions, not causal effects. The descriptive contribution of the paper does not rely on these caveats. That contribution is the picture of who scales up, when federal support arrives, and how the indirect and direct instruments differ in coverage and timing for the scale-up population.

4 Findings

4.1 Who scales up, and where?

Scale-ups are a small share of Canadian firms. Under the OECD employment definition, 6–7 percent of OECD-eligible firms qualify as scale-ups in a given year. Under the Kauffman revenue definition, 1.5–2 percent of firms qualify (Figure 1). These figures are calculated against each definition’s relevant firm population. The levels are not directly comparable across panels because the underlying populations differ.

Despite being few in number, the firms identified exert a disproportionate impact on employment and revenue (Figure 2). Under the OECD employment definition, scale-ups are about 6 percent of eligible firms but about 10 percent of employment, and under the Kauffman revenue definition, average revenue per scale-up firm is roughly twelve times the non-scale-up average. The contribution of scale-ups to aggregate Canadian employment and

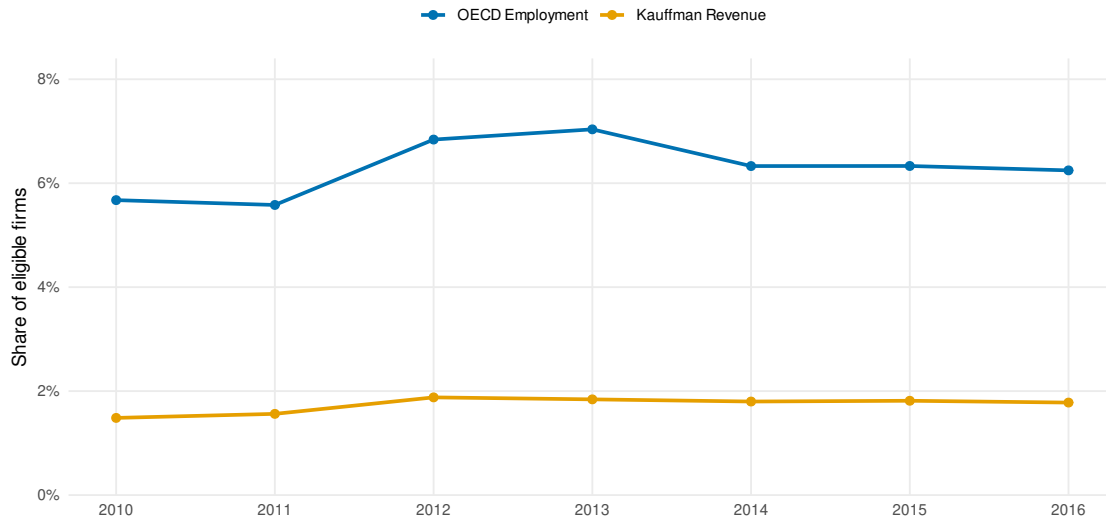


Figure 1: Scale-up incidence within definition-specific Canadian firm populations, 2010–2016. Each panel reports the annual scale-up rate for the firm population eligible under that definition. Series are Canadian administrative data, not raw OECD or Kauffman series. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.

output is substantial.

Geographically, scale-ups are concentrated in Canada’s largest urban Economic Regions. Under the OECD employment definition, the ten largest by scale-up count are Toronto, Lower Mainland–Southwest (Vancouver), Montréal, Calgary, Edmonton, Ottawa, Kitchener–Waterloo–Barrie, Montérégie, Hamilton–Niagara Peninsula, and Vancouver Island and Coast. These ten regions account for approximately 80 percent of scale-ups reported for Economic Regions. Toronto alone accounts for roughly 27 percent.¹ Figure 3 reports the top fifteen Economic Regions, and Figure 4 reports within-province scale-up shares for both definitions.

There are two qualifications that need to be identified. First, within-region employment scale-up rates are comparable for Canadian provinces (Figure 4). The urban clustering we observed in Figure 3 therefore reflects population concentration as much as differences in the propensity of firms to scale. Alberta records the highest within-province scale-up rate

¹Importantly, Statistics Canada disclosure rules suppress small-cell Economic Regions; the percentages reported here are shares of scale-ups in Economic Regions that were vetted and approved for release in the 2019 vetting cycle. These are not shares of all Canadian scale-ups.



Figure 2: Economic footprint of Canadian scale-ups, 2016. Panel A: scale-ups' share of all firms, of total employment, and of total revenue, by definition. Panel B: relative average revenue per firm, normalized to non-scale-ups = 1. Source: Authors' calculations based on linked Statistics Canada administrative firm-level data.

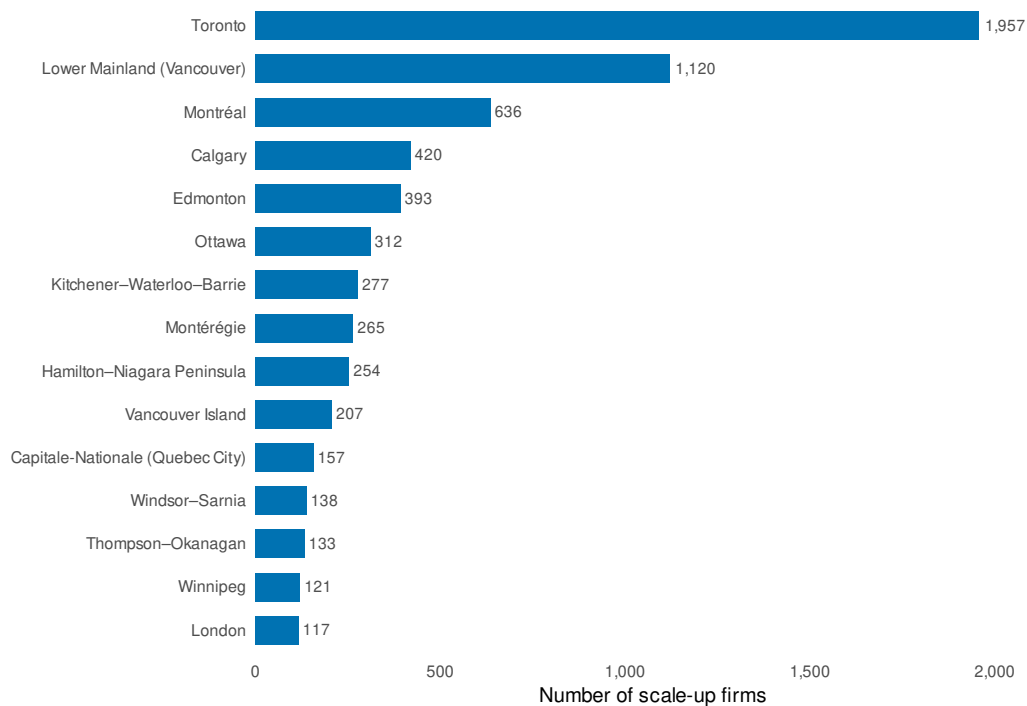


Figure 3: Top 15 Canadian Economic Regions by scale-up firm count, OECD employment definition, pooled 2011–2016. Economic Region is the Statistics Canada classification closest to a metropolitan-area concept and is the geographic unit at which the linked records are coded. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.

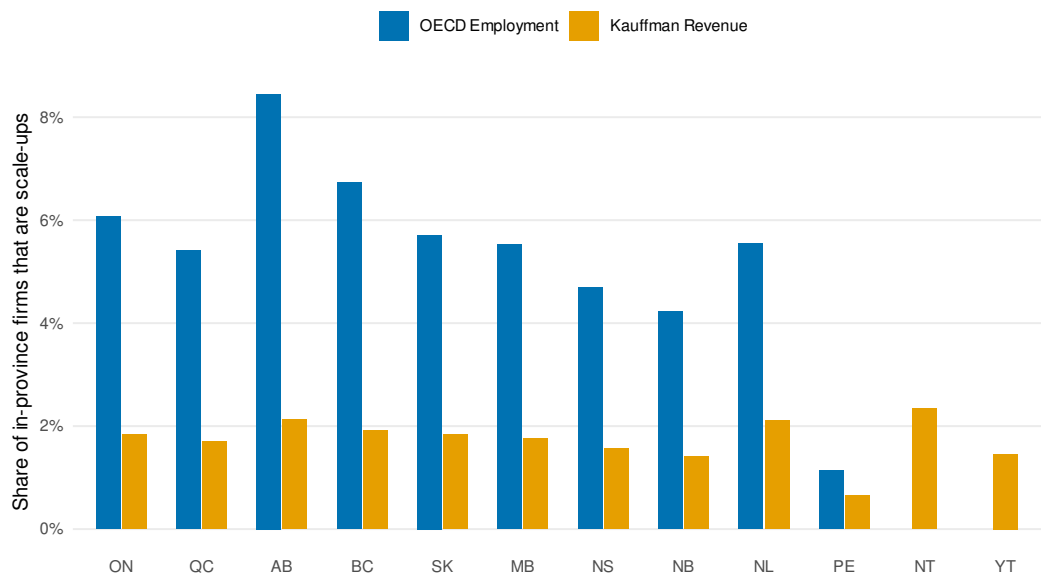


Figure 4: Within-province share of firms classified as scale-ups, by definition, pooled 2011–2016. OECD employment data for Northwest Territories (NT) and Yukon (YT) are absent because too few firms in those territories meet the 10-employee minimum to clear Statistics Canada disclosure thresholds; the Kauffman revenue definition has no analogous size minimum and so is reported for both territories. The two definitions apply to different underlying firm populations; levels are not directly comparable. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.

under the OECD employment definition (8.4 percent), driven in part by the strong 2012–2013 episode visible in Appendix Figure A.3. Second, the regional distribution of scale-ups shifted meaningfully over the panel. Under the OECD employment definition, the Prairie Provinces’ share rose sharply through 2013 and then declined, while British Columbia’s rose steadily, Ontario’s remained broadly stable, and Quebec’s declined. The Prairie and British Columbia patterns hold under the Kauffman revenue definition as well, while Quebec’s share is flat rather than declining (Appendix Figure A.4).

The industry distribution of Canadian scale-ups is broad. Only about 8–9 percent operate in ICT or directly related sectors. The largest shares are in professional, scientific, and technical services, in wholesale and retail, in construction, and in selected manufacturing industries (Figure 5). Sectoral differences clearly matter for how firms scale. Accommodation and food services and retail scale-ups tend to be employment-rich but less productive, while tech and finance scale-ups achieve revenue and productivity growth with fewer new jobs. Scale-ups come in different “types” (employment-intensive versus productivity-intensive).

Canadian scale-ups also differ from the average Canadian firm in wages, innovation, and international orientation (Tables 1 and 2). In 2016, 8 to 12 percent of scale-ups reported R&D expenditures and 17 to 18 percent were exporters, depending on the definition. The corresponding non-scale-up shares are 1 to 6 percent for R&D and 3 to 15 percent for exporting. The gap is proportionally widest under the Kauffman revenue definition, where scale-ups are about seven times as likely to report R&D and about six times as likely to export.

4.2 Federal innovation support and the scale-up population

We now turn to the question of how the federal innovation policy mix reaches Canadian scale-ups, when in their scaling trajectory it does so, and how the indirect (SR&ED) and direct (BIGS) instruments differ in coverage and timing.

Coverage is the first notable finding. Most scale-ups do not receive federal innovation

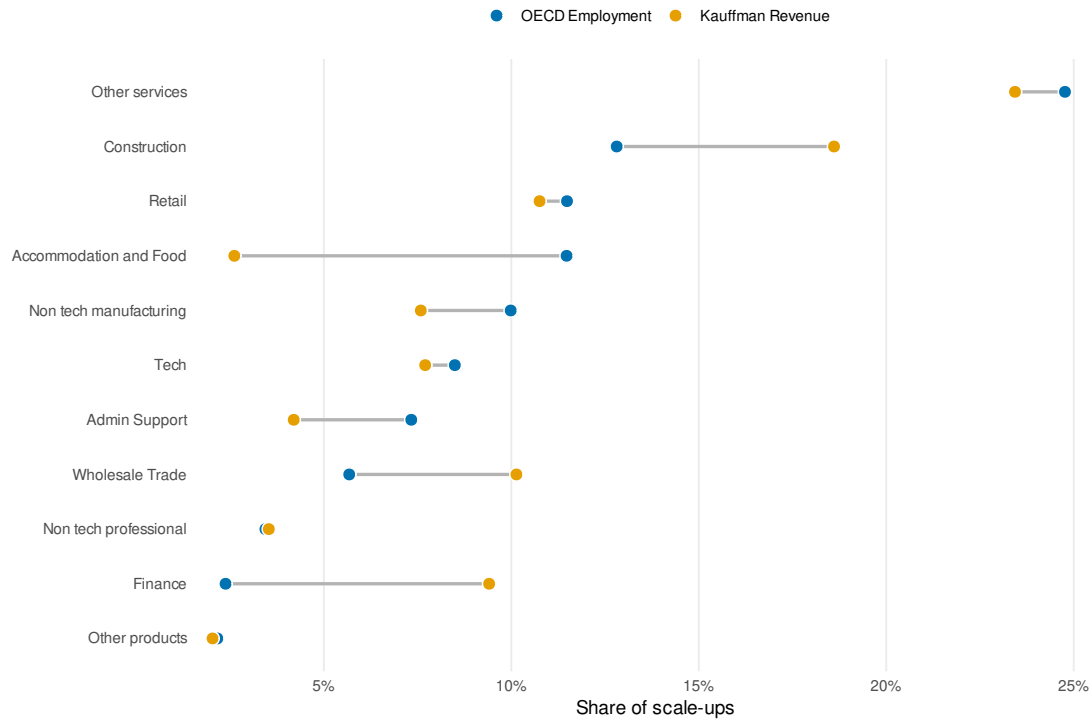


Figure 5: Distribution of Canadian scale-ups across industries, 2016. Values represent the percentage of all scale-up firms in each industry under the OECD employment and Kauffman revenue definitions. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.

	Non-scale-ups	Scale-ups
<i>Average pay (CAD)</i>		
OECD Employment	52,400	49,600
Kauffman Revenue	55,600	64,000
<i>Average productivity growth</i>		
OECD Employment	−1.67%	+2.17%
Kauffman Revenue	−2.69%	+8.78%

Table 1: Wages and productivity, scale-ups vs. non-scale-ups, 2016. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.

	Non-scale-ups	Scale-ups
<i>Share of exporters</i>		
OECD Employment	14.60%	18.37%
Kauffman Revenue	2.62%	17.28%
<i>Share of R&D spenders</i>		
OECD Employment	6.18%	11.55%
Kauffman Revenue	1.14%	8.01%

Table 2: Innovation and internationalization, scale-ups vs. non-scale-ups, 2016. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.

support. In 2019, only 7–11 percent of scale-ups had accessed BIGS or SR&ED programming. Following the 2012–2015 scale-up cohort over 2007–2019, cumulative participation rises to 15–23 percent (Figure 6). Roughly three-quarters of scale-ups grew without direct federal assistance within our panel. The BIGS and SR&ED programs are supporting only a minority of the scale-up population overall.

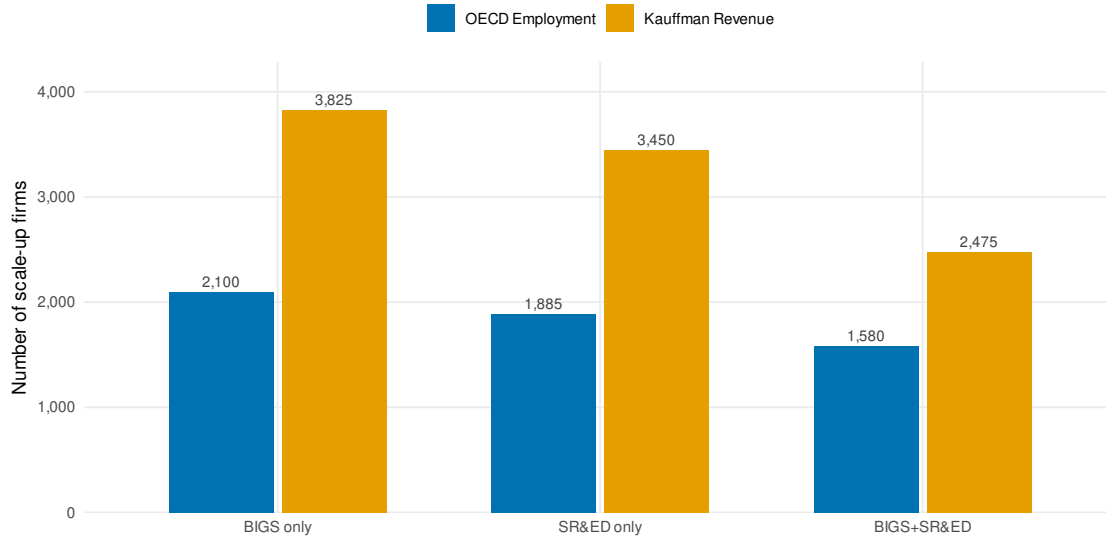


Figure 6: Support status of Canadian scale-ups in 2019, by definition (2010–2019 firm cohort). “No Support” represents firms that received neither BIGS nor SR&ED; “BIGS only” and “SR&ED only” represent exclusive recipients; “BIGS+SR&ED” represents firms receiving both. The dominant No-Support category is omitted from the plot for scale; coverage shares are reported in the text. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.

Within the supported minority, uptake concentrates among innovation-intensive scale-

ups. Some 72–77 percent of patenting scale-ups and 54–57 percent of R&D performers accessed BIGS at some point during the panel (Figure 7). Among large innovative scale-ups (100+ employees) the share reaches 85–86 and 69–70 percent, depending on scale-up type. Direct support disproportionately flows to firms already on the innovation pathway and further along the growth trajectory. A notable minority of BIGS recipients (approximately 20–30 percent, program- and definition-specific) had already scaled before first BIGS receipt (Appendix Figure A.1). Appendix Figure A.2 reports the corresponding distribution of supported scale-ups by firm age at first scaling.

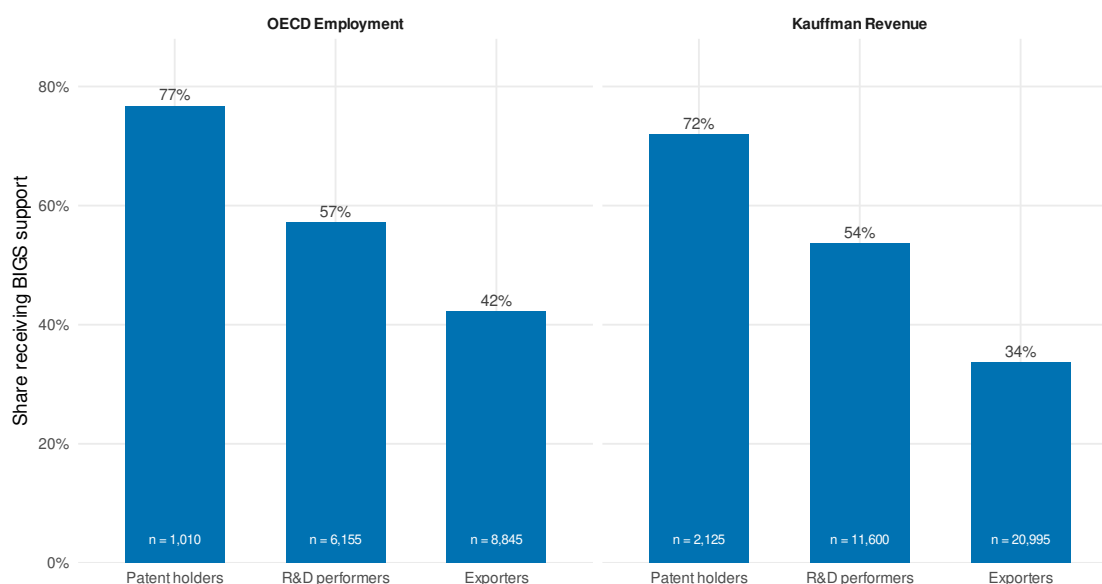


Figure 7: BIGS participation among innovative scale-ups, 2010–2015 cohort. Bars report the share of each group that received BIGS support; the label at each bar’s base gives the group total. A firm is classified as an exporter or R&D performer if it reported exports or R&D expenditures in any year during 2007–2019, and as a patenter if it filed at least one patent application during 2007–2015. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.

The direct and indirect instruments differ in the stage at which they reach scale-ups. More than half of BIGS-supported scale-ups first received direct funding after their initial scaling episode. Approximately 86 percent of SR&ED-supported scale-ups, by contrast, claimed credits before or during scaling (Figure 8). This divergence reflects the different structures of the two programs. SR&ED is a tax credit tied to qualifying R&D expenditures

and is therefore available to firms already conducting R&D. Consequently, indirect support often precedes or coincides with scaling. BIGS programs, on the other hand, use application-and-adjudication processes that tend to favor firms with demonstrated capacity. Funding therefore often arrives after a firm has begun to scale.

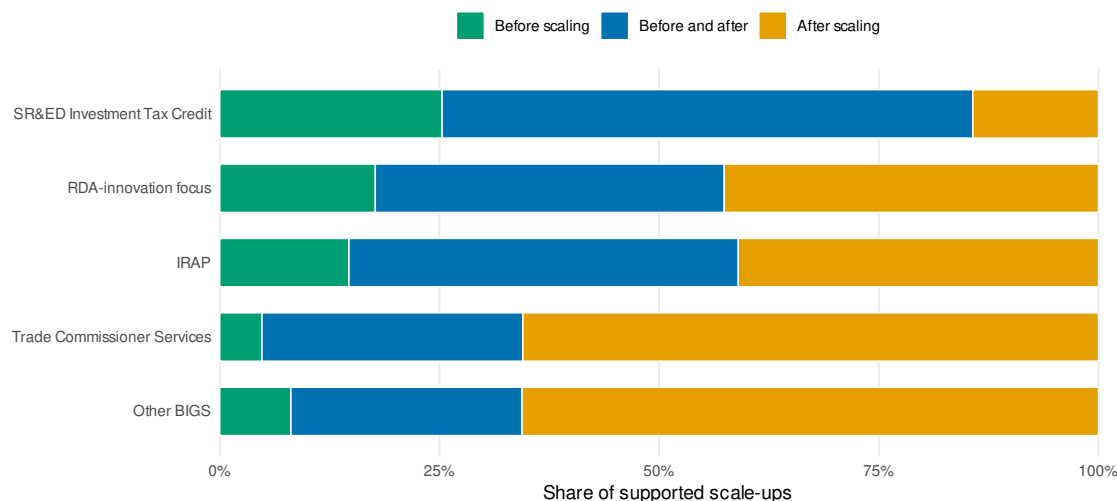


Figure 8: Timing of first receipt of BIGS and SR&ED support relative to initial scaling for the 2012–2015 cohort, using the OECD employment definition. Firms are categorized according to whether they received support only before scaling, both before and after scaling, or only after scaling. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.

The dollar-volume finding (Figure 9) supports the timing story. The bulk of both BIGS and SR&ED dollars flows to firms post-scaling. This partly reflects program mechanics, as larger firms with higher eligible expenditures qualify for larger SR&ED claims and BIGS programs channel funding to more established firms. It also reflects firm-side selection, as companies intensify R&D and financing activity once growth is underway. These findings suggest an important pattern: federal innovation policy plays more of a reinforcing than an initiating role for the scale-up population. Most support flows to firms after they are established on an upward growth trajectory rather than before scaling begins. In other words, support flows to market-tested firms.

BIGS dollars are also concentrated by sector and firm size (Figure 10). About 70 percent of BIGS value to scale-ups flows to manufacturing, information and cultural industries, and

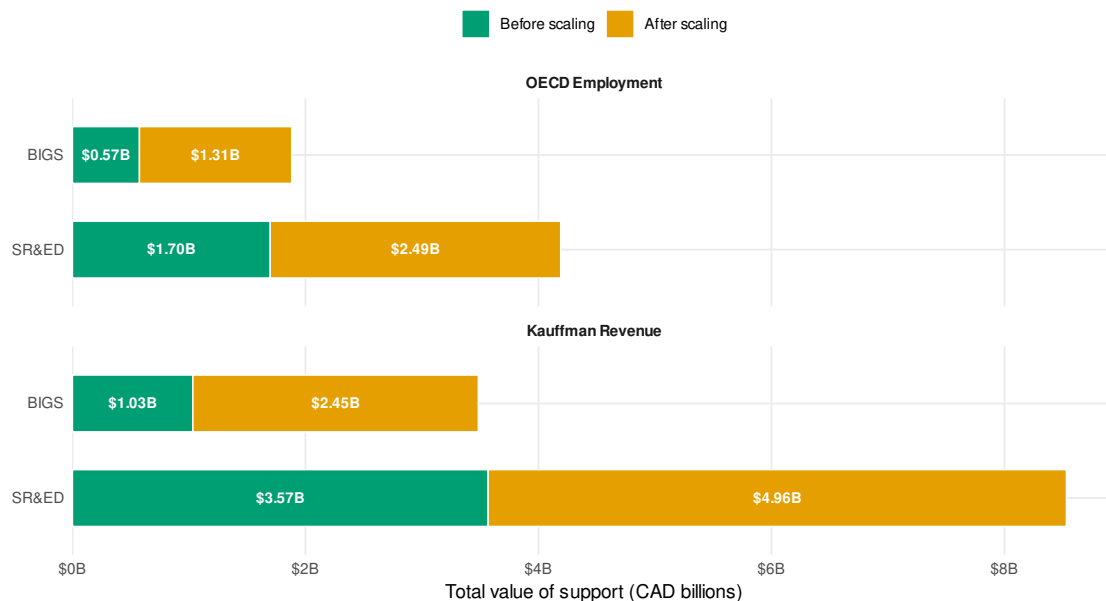


Figure 9: Total value of BIGS and SR&ED support, 2007–2019, by timing relative to initial scaling, 2012–2015 cohort. BIGS and SR&ED values shown separately for revenue-defined and employment-defined scale-ups. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.

professional, scientific, and technical services. Around 60 percent goes to firms with more than 50 employees. Technology-intensive industries receive, on average, twice as much BIGS funding per recipient firm as non-technology sectors. IRAP and the Trade Commissioner Service are the most common touchpoints among scale-ups with smaller average award sizes, while the Strategic Innovation Fund and Sustainable Development Technologies Canada are less common but provide much higher per-firm value (Appendix Table A.1).

4.3 Conditional differences in outcomes associated with BIGS and SR&ED

We now turn to the matched-cohort regressions, which we read as conditional descriptive associations under stated assumptions about selection on observables, not causal effects of the programs. For BIGS, Table 3 reports estimates under both scale-up definitions. The conditional difference in subsequent employment is 0.10 log points under the OECD employment definition and 0.14 under the Kauffman revenue definition, corresponding to approximately 10 and 14 percent higher employment among BIGS recipients relative to matched

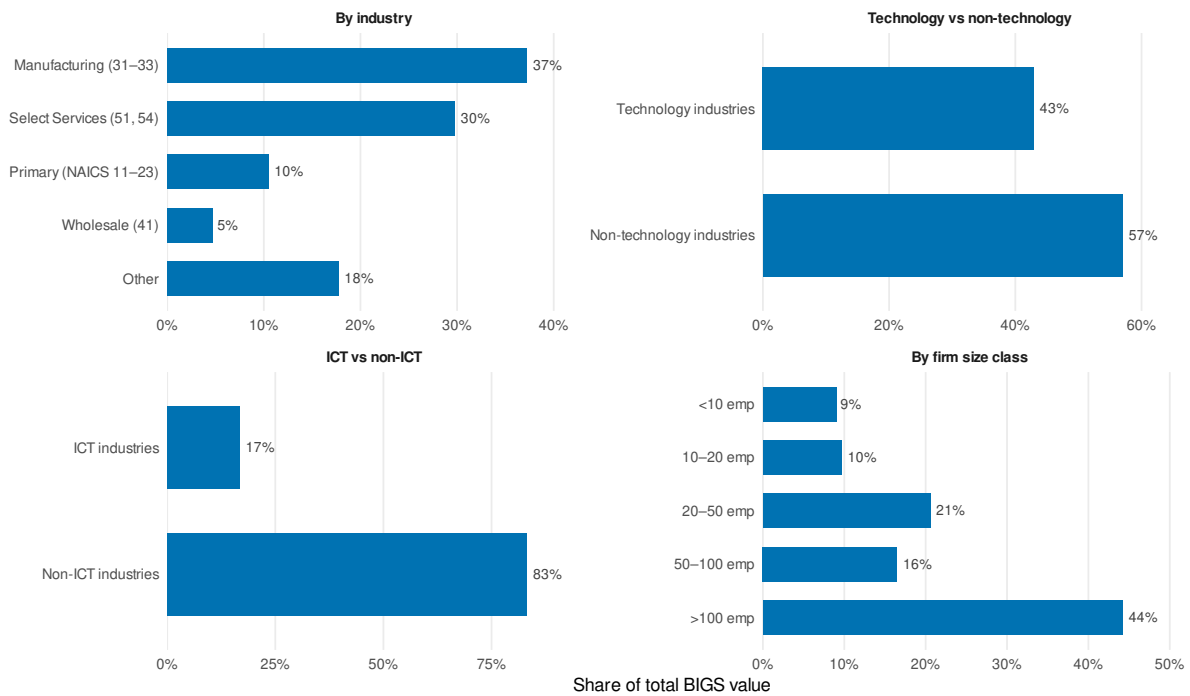


Figure 10: BIGS support value distribution for 2019 scale-ups, 2010–2019 cohort, Kauffman revenue definition. Industry classification follows the four-digit NAICS codes with at least three times the national average concentration of technology workers. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.

non-recipients. For revenue, the corresponding estimates are 0.17 and 0.20 log points, or approximately 17 and 20 percent higher revenue. Differences in R&D expenditures, exports, and wages are also positive for both definitions. The productivity (sales per employee) difference is not significantly different from zero in the near term, suggesting that the stronger employment and revenue outcomes among BIGS recipients do not translate into a measurable productivity difference over that same period.

The estimates are robust across the dimensions we can assess. Matching improves covariate balance (Kolmogorov–Smirnov tests), and visual inspection of pre-trends does not show sustained pre-treatment divergence. Estimates are also robust to alternative calipers, matching ratios, and staggered DiD specifications (Callaway and Sant’Anna, 2021). We control for SR&ED in the BIGS regression and for BIGS in the SR&ED specification, although overlap in program use may attenuate program-specific estimates. The analysis also does not capture the full range of firm outcomes. Service exports and procurement effects are not fully captured, while the support definition excludes the Business Development Bank of Canada, Export Development Canada, and provincial-level programs.

SR&ED is a control in the BIGS treatment model, but its relationship to subsequent growth requires separate attention. Table 4 reports pooled-OLS estimates for the full firm-year panel, controlling for lagged outcomes, firm age, industry, province, and year fixed effects. Unlike the matched-cohort analysis of scale-ups in Table 3, these estimates describe associations across the full firm population. This difference in design reflects the timing of program participation. BIGS typically reaches firms after they have scaled, making the matched-cohort design we used here feasible, whereas SR&ED is frequently used before or during scaling (Figures 8 and 9). A comparable post-scaling design is therefore not well suited to SR&ED, as restricting the analysis to firms that have already scaled would shift the windows of observation beyond the period when many firms first use indirect support.

The SR&ED estimates are positive for both employment and revenue. For employment, the trigger and switch estimates are 0.105 and 0.172, respectively, corresponding approxi-

Outcome	OECD Employment		Kauffman Revenue	
	Estimate	Sig.	Estimate	Sig.
asinh(Employment)	0.100	***	0.140	***
asinh(Revenue)	0.170	***	0.200	***
Scaling incidence (LPM)	0.050	***	0.040	***
asinh(Average wages)	0.020	*	0.030	***
asinh(R&D expenditures)	0.150	***	0.140	***
R&D incidence (LPM)	0.050	***	0.050	***
asinh(Exports)	0.170	**	0.120	**
Export incidence (LPM)	0.040	***	0.050	***
asinh(Sales per employee)	0.020		0.010	
Firm FE	Yes		Yes	
Industry $\times$ Year FE	Yes		Yes	
Lagged-outcome match	Yes		Yes	
SR&ED control	Yes		Yes	

Table 3: Conditional differences in scale-up performance associated with BIGS receipt, matched-cohort fixed-effects estimates, 2012–2014 treated cohort. Estimates from equation (1). Sample window: $t - 5$ to $t + 5$ around first BIGS receipt. Matched control firms selected via propensity-score matching on industry, size class, age, R&D performer, exporter, patent holder, and lagged outcomes. Results are reported in accordance with Statistics Canada confidentiality and disclosure-control requirements. Finer-grained disclosure of clustered standard errors, exact sample sizes, and R^2 statistics was not requested. Coefficients on asinh-transformed outcomes are interpretable as proportional differences. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Treatment definition	Employment outcome		Revenue outcome	
	Coef.	SE	Coef.	SE
SR&ED trigger (first year of R&D expenditure)	0.105***	(0.001)	0.735***	(0.005)
SR&ED switch (contemporaneous R&D)	0.172***	(0.001)	1.128***	(0.004)
Industry / Province / Year FE	Yes		Yes	
Firm age, lagged outcome	Yes		Yes	
N (firm-year obs., emp. outcome)	14,125,278		—	
N (firm-year obs., rev. outcome)	—		22,858,380	

Table 4: Pooled-OLS association between SR&ED participation and subsequent growth, full firm-year panel. Descriptive; not comparable in design to the matched-cohort estimates in Table 3 (see text). Lagged outcome (asinh), firm age, two-digit NAICS industry, province, and year fixed effects. Standard errors, clustered at the firm level, in parentheses. Outcomes are asinh-transformed; coefficients are interpretable as approximate proportional differences when small. “Trigger” is one from the firm’s first year with qualifying R&D expenditures and one thereafter. “Switch” is one in years with positive R&D expenditures. The SR&ED indicator includes firms claiming SR&ED deductions, not only firms receiving investment tax credits. Results are reported in accordance with Statistics Canada confidentiality and disclosure-control requirements. Significance: *** $p < 0.01$.

mately to 11 and 17 percent higher employment. The corresponding revenue estimates are considerably larger, at 0.735 and 1.128, indicating substantially higher revenue among firms that begin reporting qualifying R&D expenditures and those with R&D activity in a given year. Note that these pooled-OLS estimates use the full firm-year panel rather than the scale-up population and are not directly comparable to the scale-up-focused matched-cohort results in Table 3. They answer a related but distinct question about whether SR&ED participation is associated with subsequent growth among Canadian firms generally.

The estimates in Table 4 should be interpreted with the same caution as Table 3. The selection-on-observables limitations discussed in Section 3 apply. An additional consideration specific to the SR&ED specification is that treatment is defined by the onset of qualifying R&D expenditure, which necessarily conditions on a firm becoming an R&D performer, itself a selected behavior that likely reflects underlying innovative capability, growth potential, and orientation. The estimates are best read as the conditional difference in subsequent outcomes among firms that transition into R&D-performing status, not as the effect of an additional dollar of SR&ED support. A matched or quasi-experimental SR&ED-as-treatment specification restricted to the scale-up population, comparable in design to Table 3, can therefore be seen as a priority for future work.

Program-level results are reported in Appendix Table A.1. IRAP and TCS are the most common programs used by scale-ups and provide relatively modest average awards, while the Strategic Innovation Fund and Sustainable Development Technologies Canada support fewer firms with substantially larger awards. Estimated conditional differences are broadly similar across programs and consistent with the aggregate results. However, many firms receive support from multiple programs, making it difficult to disaggregate the contribution of any one program. The results therefore provide limited evidence of meaningful differences across programs, another issue that warrants further research.

5 Discussion

The findings presented have implications for Canadian innovation policy, particularly for how federal support reaches scale-ups and when in their growth trajectories that support is provided.

Three-quarters of Canadian scale-ups do not access federal innovation support over the multi-year observation period. Federal support therefore cannot account for the growth of most scale-ups, whose growth trajectories are also shaped by firm capabilities, market demand, access to capital, and other factors. However, among firms that do receive federal support, participation is concentrated among those engaged in R&D, patenting, and exporting. Federal innovation programs thus reach a minority of scale-ups overall but a much larger share of the more innovation-intensive segment of this firm population.

The timing of support also differs across instruments. SR&ED typically reaches firms before or during scaling, whereas BIGS support more often follows the initial scaling episode. This suggests that the two instruments may play different, and potentially complementary, roles in support of growth. A reasonable inference here is that SR&ED provides support while firms are undertaking R&D, often before a rapid growth period, while direct programs tend to engage firms after they have demonstrated greater innovative and commercial capacity. Direct support may therefore enable larger or more ambitious projects, but often among firms that are market-validated and whose growth and innovative activity are already well underway.

Our findings suggest that federal direct-support programs more often reinforce existing growth rather than initiate it. There are many potential mechanisms that could reasonably produce this pattern. Firms with stronger managerial capacity and more resources may be more aware of government programs and thus more likely to seek support. Moreover, the competitive review and due-diligence requirements of many BIGS programs will likely favor firms that have already demonstrated commercial viability and the capacity to run complex

projects. Political and administrative considerations may also factor in, making support for firms with an established record of success easier to justify (Mayhew, 1974). Of course, the descriptive evidence presented here cannot distinguish among these mechanisms, which are also not mutually exclusive.

The later timing of direct support might be understood as support arriving too late, but we do not interpret it this way. Rather, we see it as highlighting a trade-off in innovation policy. Earlier intervention may help firms overcome constraints before scaling and potentially increase the number that achieve sustained growth, but identifying high-potential firms at that stage is extremely difficult. By way of comparison, firms that have already demonstrated commercial viability and innovative capability give policymakers more information on which to base funding decisions, but directing support toward these firms necessarily concentrates resources among those already on a growth trajectory.

The policy design implications follow from this trade-off. Policymakers must balance support for promising firms earlier in their development with support for firms that have already demonstrated the capacity to contribute to innovation, productivity growth, and international competitiveness (Harris, 2015). The issue is not simply when support is provided, but how different instruments are designed and sequenced to serve differing objectives. Given the constrained fiscal environment in which such decisions are made, this means balancing two government imperatives: helping more firms that can enter high-growth trajectories and enabling firms with demonstrated potential to continue scaling.

6 Conclusion

This paper provides the first firm-level descriptive account of how Canada’s federal innovation policy bundle interacts with the scale-up population. Four main findings emerge. First, scale-ups represent a small share of Canadian firms but account for a disproportionate share of employment and revenue. They are found across industries and regions, although they are concentrated in Canada’s largest urban Economic Regions. Second, most scale-ups

never receive federal innovation support, with participation concentrated among larger and more innovation-intensive firms. Third, the timing of support differs substantially by instrument. SR&ED typically reaches firms before or during scaling, whereas BIGS reaches them, on average, after their initial scaling episode. Fourth, among matched scale-ups, BIGS recipients exhibit stronger subsequent employment, revenue, R&D, wage, and export outcomes than comparable non-recipients, while differences in productivity are not statistically distinguishable from zero.

A main contribution of the paper is its focus on the timing and incidence of federal innovation support among firms that ultimately achieve sustained high growth. By examining support within this narrowly defined population, we document patterns in the sequencing and targeting of federal innovation instruments that would not be possible with analysis of the full population of firms or even a larger subset of them. In particular, the results suggest that direct support is concentrated among firms that have already demonstrated innovative capability and are growing. These findings provide new evidence on how Canada’s innovation support system operates, while recognizing that they may not generalize beyond the scale-up population.

While the descriptive and matched-cohort analyses do not identify the causal effects of individual programs, and unobserved differences between supported and unsupported firms may contribute to the associations we observe, the associations established are suggestive of causal relationships. We think future research could build on these findings by applying quasi-experimental designs where suitable sources of exogenous variation exist and by extending the analysis to support provided through the Business Development Bank of Canada, Export Development Canada, and provincial governments. Further work could also examine the sensitivity of the matched-cohort estimates to alternative specifications, including estimation without lagged outcomes, and develop predictive approaches to identify firms likely to become future scale-ups ([Guzman and Stern, 2017](#); [Houle and Macdonald, 2025](#)).

Canada’s federal innovation policy mix reaches a minority of scale-ups, and direct support

more often reaches firms once they are scaling, although the descriptive evidence alone cannot establish whether this timing maximizes the contribution of public support. By establishing how federal support is distributed and sequenced across firms' growth trajectories, this paper provides a foundation for assessing how innovation policy might better support the emergence and continued growth of Canadian scale-ups.

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Conflicts of interest

The authors declare no conflicts of interest.

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Data availability

The microdata underlying this paper are Statistics Canada administrative records accessible only inside the Statistics Canada secure-access environment under approved research-license terms. A replication package containing disclosed aggregations, codebooks, and R scripts

that reproduce the figures from those disclosed aggregations is available from the authors and will be archived alongside the published version of the paper.

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Appendix

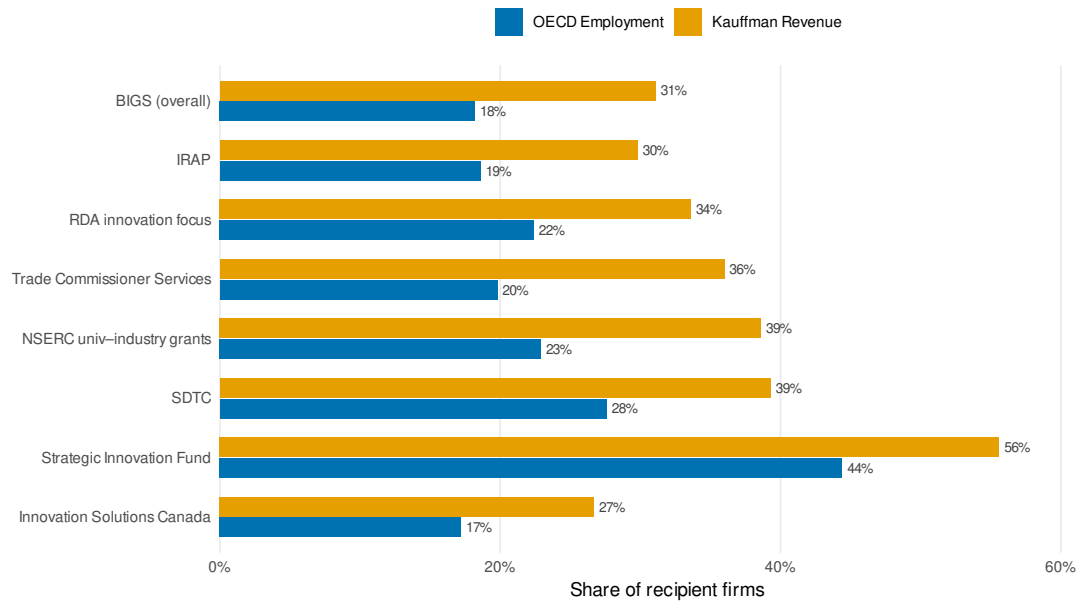


Figure A.1: Share of BIGS recipients already scaled before first receipt, by program and definition. Source: Authors' calculations based on linked Statistics Canada administrative firm-level data.

Program	Share of recipients	Average award value
Industrial Research Assistance Program (IRAP)	High	Low–Medium
Trade Commissioner Service (TCS)	High	Low
Regional Development Agencies (RDA, innovation focus)	Medium	Medium
Other BIGS programs	Medium	Low–Medium
Strategic Innovation Fund (SIF)	Low	High
Sustainable Development Technologies Canada (SDTC)	Low	High

Table A.1: Program-level breakdown of BIGS support among scale-ups, 2010–2019. Programs ordered by share of recipient scale-ups. Source: Authors' calculations based on linked Statistics Canada administrative firm-level data. Categories use ordinal classifications because dollar-value cells fall below Statistics Canada disclosure thresholds for some program-cohort cells.

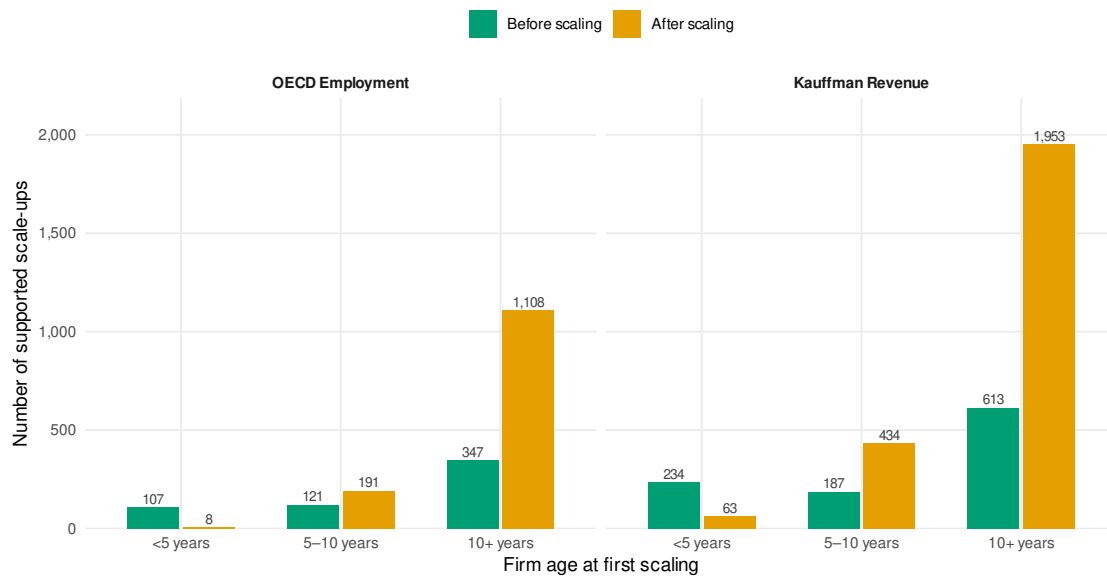


Figure A.2: Number of BIGS-supported scale-ups by firm age at first scaling, OECD employment and Kauffman revenue definitions, 2012–2015 cohort. Bars distinguish firms whose first BIGS receipt preceded initial scaling from those whose first receipt followed it. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.

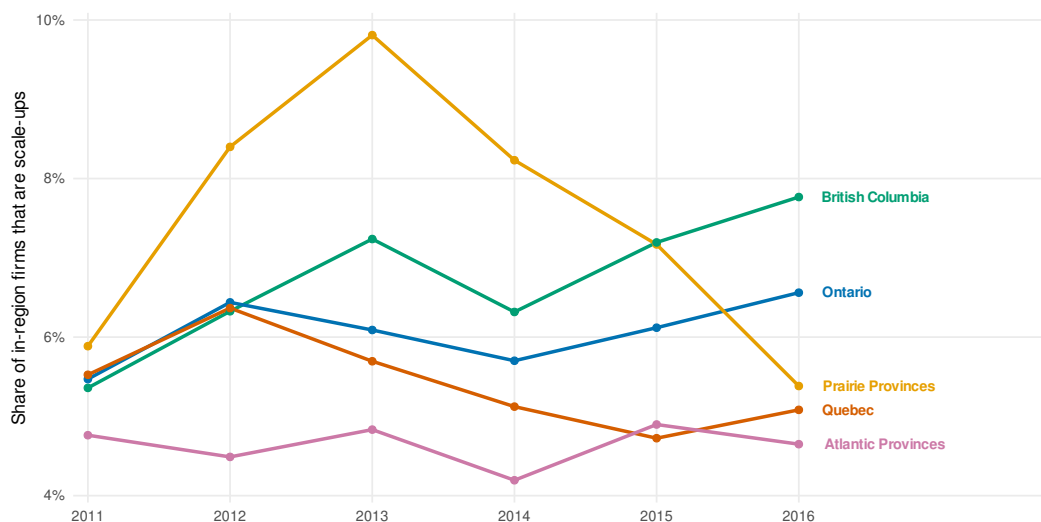


Figure A.3: Within-region scale-up share by Canadian region, OECD employment definition, 2011–2016. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.

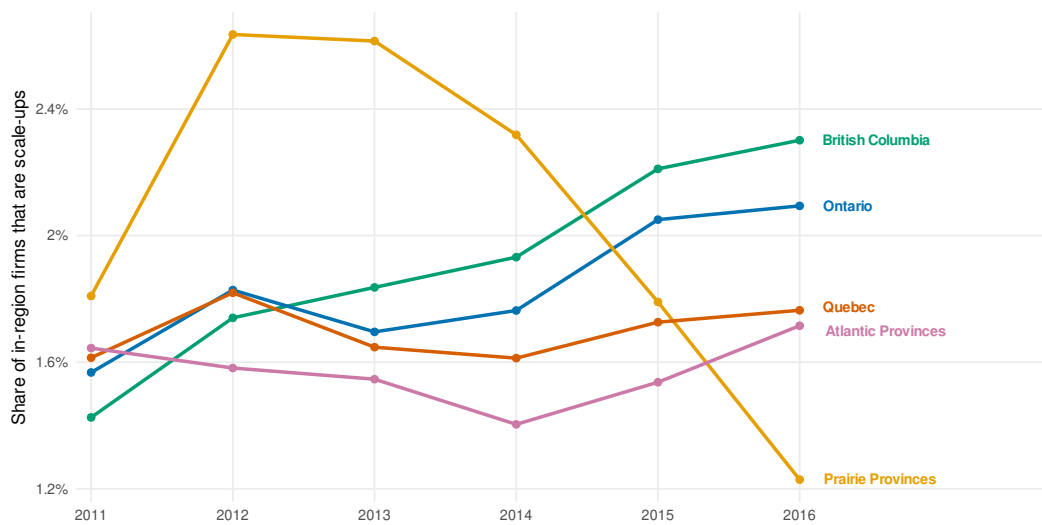


Figure A.4: Within-region scale-up share by Canadian region, Kauffman revenue definition, 2011–2016. The northern territories are omitted because the disclosed series covers only 2012–2015 there. Source: Authors’ calculations based on linked Statistics Canada administrative firm-level data.